Project 1

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DSC 680: Applied Data Science

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Import Libraries, read data, and preview dataframe

In [1]:
```
import pandas as pd
import numpy
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn as skl
from tqdm import tqdm
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
```

In [2]:
```
import os
print(os.getcwd())
```

```
C:\Users\velaz\Desktop\JupyterProject
```

In [3]:
```
# Read File and create data frame
file_path = "customer_segmentation.csv"
df = pd.read_csv(file_path)
print(df.head())
```

```
     ID  Year_Birth   Education Marital_Status   Income  Kidhome  Teenhome  \
0  5524        1957  Graduation         Single  58138.0        0         0
1  2174        1954  Graduation         Single  46344.0        1         1
2  4141        1965  Graduation       Together  71613.0        0         0
3  6182        1984  Graduation       Together  26646.0        1         0
4  5324        1981         PhD        Married  58293.0        1         0

  Dt_Customer  Recency  MntWines  ...  NumWebVisitsMonth  AcceptedCmp3  \
0  04-09-2012       58       635  ...                  7             0
1  08-03-2014       38        11  ...                  5             0
2  21-08-2013       26       426  ...                  4             0
3  10-02-2014       26        11  ...                  6             0
4  19-01-2014       94       173  ...                  5             0

   AcceptedCmp4  AcceptedCmp5  AcceptedCmp1  AcceptedCmp2  Complain  \
0             0             0             0             0         0
1             0             0             0             0         0
2             0             0             0             0         0
3             0             0             0             0         0
4             0             0             0             0         0

   Z_CostContact  Z_Revenue  Response
0              3         11         1
1              3         11         0
2              3         11         0
3              3         11         0
4              3         11         0

[5 rows x 29 columns]
```

In [4]:
```
# Data Cleaning and Feature ENgineering
```

In [5]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 29 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   ID                   2240 non-null   int64
 1   Year_Birth           2240 non-null   int64
 2   Education            2240 non-null   str
 3   Marital_Status       2240 non-null   str
 4   Income               2216 non-null   float64
 5   Kidhome              2240 non-null   int64
 6   Teenhome             2240 non-null   int64
 7   Dt_Customer          2240 non-null   str
 8   Recency              2240 non-null   int64
 9   MntWines             2240 non-null   int64
 10  MntFruits            2240 non-null   int64
 11  MntMeatProducts      2240 non-null   int64
 12  MntFishProducts      2240 non-null   int64
 13  MntSweetProducts     2240 non-null   int64
 14  MntGoldProds         2240 non-null   int64
 15  NumDealsPurchases    2240 non-null   int64
 16  NumWebPurchases      2240 non-null   int64
 17  NumCatalogPurchases  2240 non-null   int64
 18  NumStorePurchases    2240 non-null   int64
 19  NumWebVisitsMonth    2240 non-null   int64
 20  AcceptedCmp3         2240 non-null   int64
 21  AcceptedCmp4         2240 non-null   int64
 22  AcceptedCmp5         2240 non-null   int64
 23  AcceptedCmp1         2240 non-null   int64
 24  AcceptedCmp2         2240 non-null   int64
 25  Complain             2240 non-null   int64
 26  Z_CostContact        2240 non-null   int64
 27  Z_Revenue            2240 non-null   int64
 28  Response             2240 non-null   int64
dtypes: float64(1), int64(25), str(3)
memory usage: 507.6 KB
```

In [6]:
```
df = df.dropna()
```

In [7]:
```
# Investigate the values
df['Marital_Status'].value_counts()
```

Out[7]:
```
Marital_Status
Married     857
Together    573
Single      471
Divorced    232
Widow        76
Alone         3
Absurd        2
YOLO          2
Name: count, dtype: int64
```

In [8]:
```
df['Education'].value_counts()
```

Out[8]:
```
Education
Graduation    1116
PhD            481
Master         365
2n Cycle       200
Basic           54
Name: count, dtype: int64
```

In [9]:
```
df['Dt_Customer'].value_counts()
```

Out[9]:
```
Dt_Customer
31-08-2012    12
12-05-2014    11
12-09-2012    11
14-02-2013    11
22-05-2014    10
              ..
01-11-2013     1
10-08-2013     1
11-10-2012     1
20-12-2012     1
09-01-2014     1
Name: count, Length: 662, dtype: int64
```

In [10]:
```
df['Dt_Customer'] = pd.to_datetime(df['Dt_Customer'], dayfirst = True)
```

In [11]:
```
df['Dt_Customer']
```

Out[11]:
```
0      2012-09-04
1      2014-03-08
2      2013-08-21
3      2014-02-10
4      2014-01-19
          ...
2235   2013-06-13
2236   2014-06-10
2237   2014-01-25
2238   2014-01-24
2239   2012-10-15
Name: Dt_Customer, Length: 2216, dtype: datetime64[us]
```

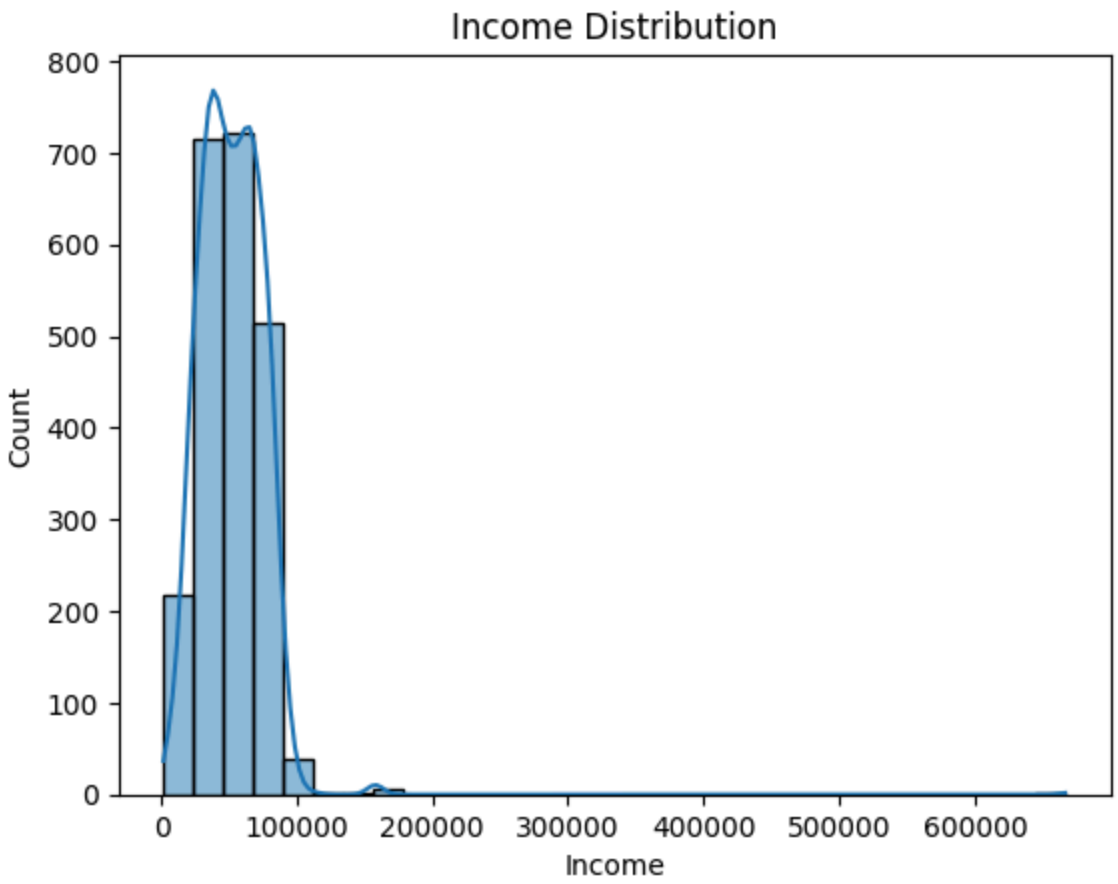
In [12]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 2216 entries, 0 to 2239
Data columns (total 29 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   ID                   2216 non-null   int64
 1   Year_Birth           2216 non-null   int64
 2   Education            2216 non-null   str
 3   Marital_Status       2216 non-null   str
 4   Income               2216 non-null   float64
 5   Kidhome              2216 non-null   int64
 6   Teenhome             2216 non-null   int64
 7   Dt_Customer          2216 non-null   datetime64[us]
 8   Recency              2216 non-null   int64
 9   MntWines             2216 non-null   int64
 10  MntFruits            2216 non-null   int64
 11  MntMeatProducts      2216 non-null   int64
 12  MntFishProducts      2216 non-null   int64
 13  MntSweetProducts     2216 non-null   int64
 14  MntGoldProds         2216 non-null   int64
 15  NumDealsPurchases    2216 non-null   int64
 16  NumWebPurchases      2216 non-null   int64
 17  NumCatalogPurchases  2216 non-null   int64
 18  NumStorePurchases    2216 non-null   int64
 19  NumWebVisitsMonth    2216 non-null   int64
 20  AcceptedCmp3         2216 non-null   int64
 21  AcceptedCmp4         2216 non-null   int64
 22  AcceptedCmp5         2216 non-null   int64
 23  AcceptedCmp1         2216 non-null   int64
 24  AcceptedCmp2         2216 non-null   int64
 25  Complain             2216 non-null   int64
 26  Z_CostContact        2216 non-null   int64
 27  Z_Revenue            2216 non-null   int64
 28  Response             2216 non-null   int64
dtypes: datetime64[us](1), float64(1), int64(25), str(2)
memory usage: 519.4 KB
```

In [13]:
```
# Make some quick overview visuals
```

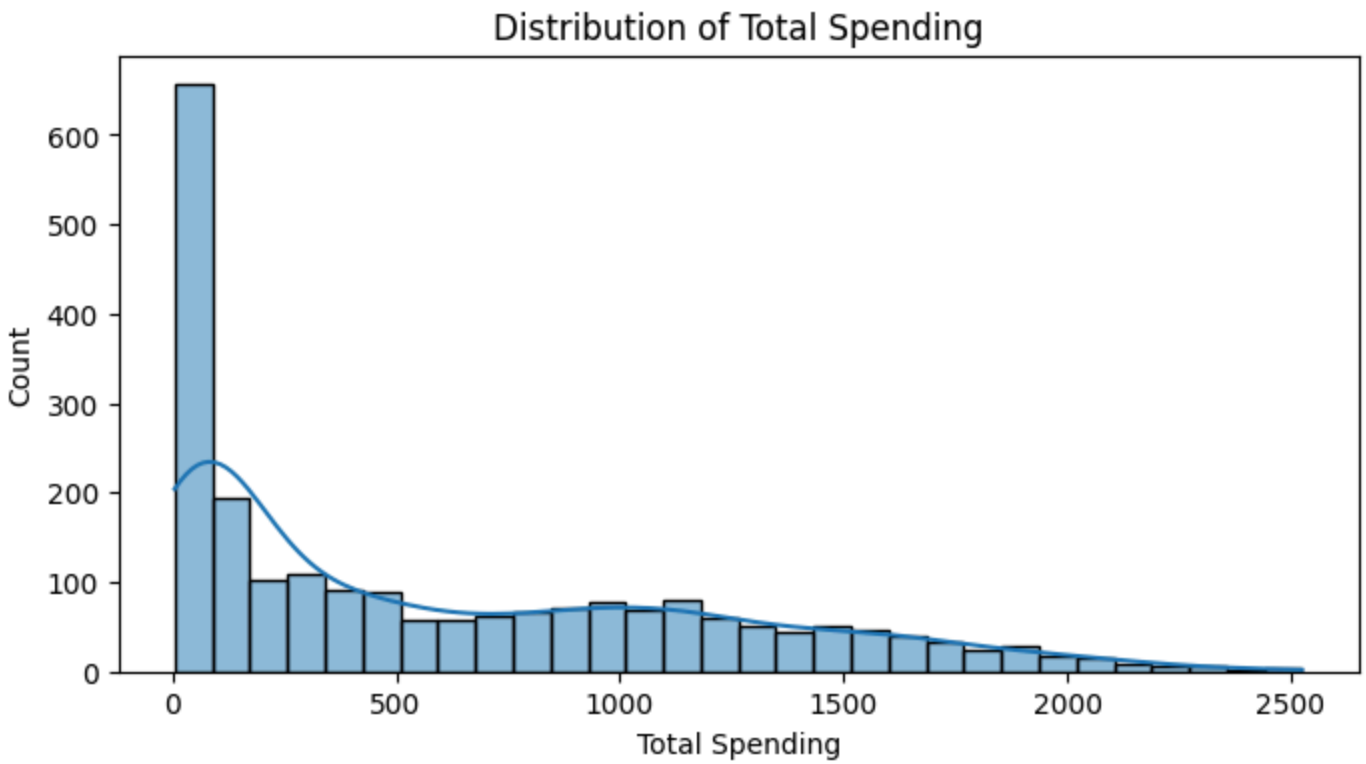
In [14]:
```
## Income Distribution
sns.histplot(df['Income'], bins = 30, kde=True)
```

```
plt.title('Income Distribution')
plt.show()
```



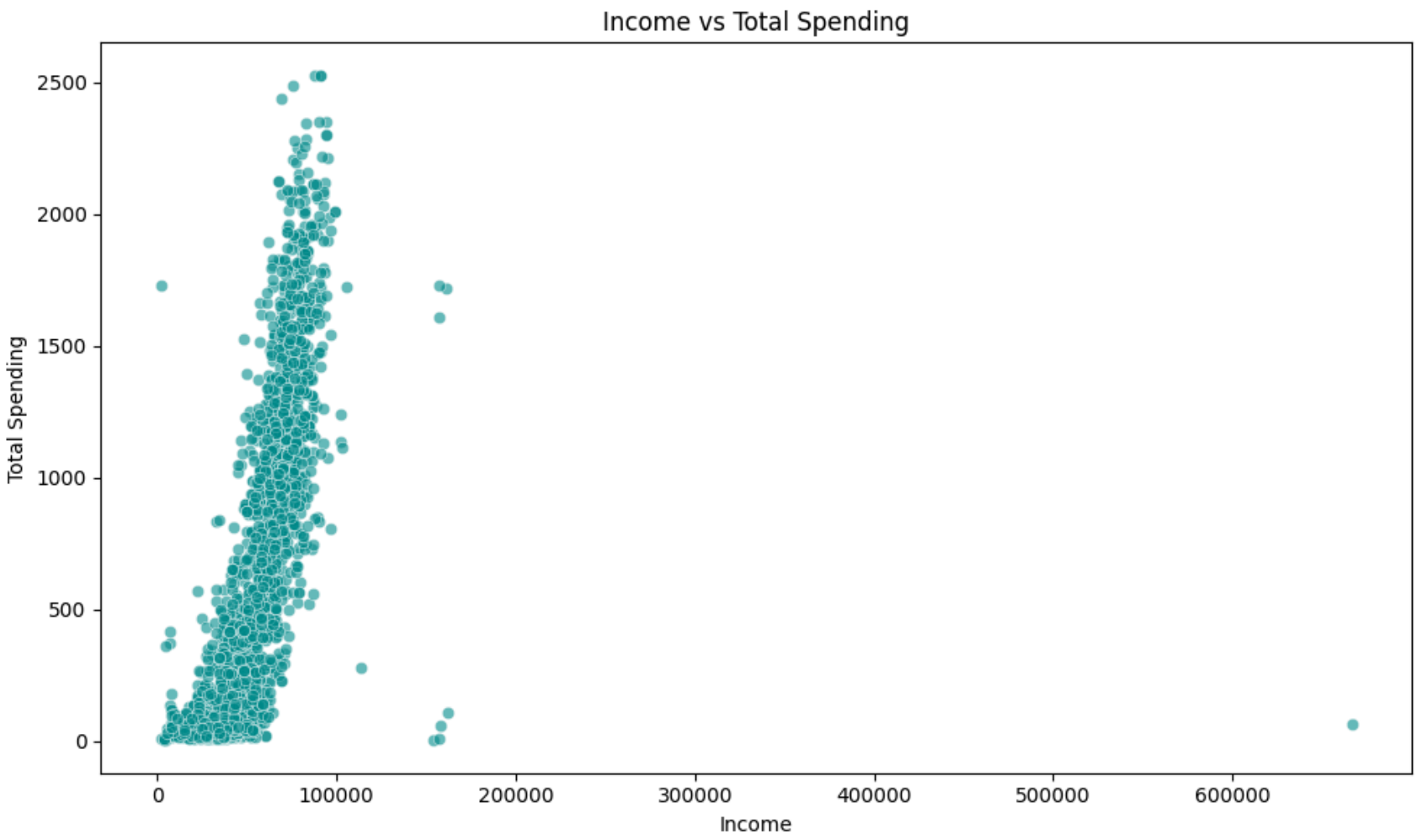
In [15]:
```
## Total Spending Distribution
spend_cols = ['MntWines','MntFruits','MntMeatProducts','MntFishProducts','MntSweetProducts','MntGoldProds']
purchase_cols = [
    "NumWebPurchases",
    "NumCatalogPurchases",
    "NumStorePurchases"
]
df['Total_Spending'] = df[spend_cols].sum(axis=1)
df["Total_Purchases"] = df[purchase_cols].sum(axis=1)
plt.figure(figsize=(8,4))
sns.histplot(df['Total_Spending'], bins=30, kde=True)
```

```
plt.title('Distribution of Total Spending')
plt.xlabel('Total Spending')
plt.ylabel('Count')
plt.show()
```



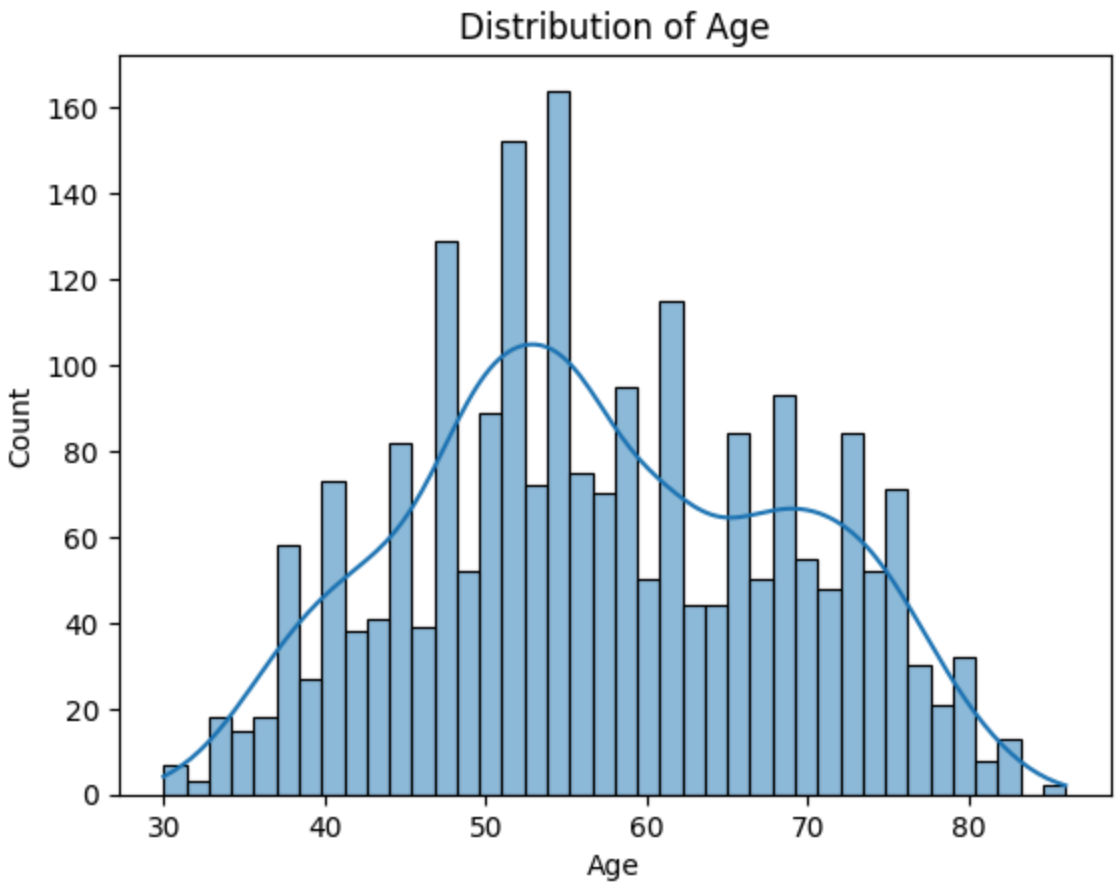
In [16]:
```
# Income vs Total Spending
plt.figure(figsize=(10, 6))
sns.scatterplot(
    data= df,
    x="Income",
    y="Total_Spending",
    alpha=0.6,
    color="darkcyan"
)
plt.title("Income vs Total Spending")
plt.xlabel("Income")
plt.ylabel("Total Spending")
```

```
plt.tight_layout()
plt.show()
```



In [17]:
```
## Age Distribution
df["Age"] = 2026- df['Year_Birth']
# Remove unrealistic ages
df = df[(df["Age"] >= 18) & (df["Age"] <= 100)].copy()
sns.histplot(df['Age'], bins=40, kde=True)
plt.title('Distribution of Age')
plt.xlabel('Age')
plt.ylabel('Count')
```

Out[17]: Text(0, 0.5, 'Count')



In [18]:
```
## New Column
df['Customer_Since'] = (pd.Timestamp('today') - df['Dt_Customer']).dt.days
```

In [19]:
```
## New Column
df["Total_Children"] = df['Kidhome'] + df['Teenhome']
```

In [20]:
```
# Insight from the columns
summary_cols = [
    "Income",
    "Age",
    "Recency",
    "Total_Spending",
```

```
    "Total_Purchases",
    "NumWebVisitsMonth",
    "Total_Children",
    "Response"
]

summary_stats = df[summary_cols].describe().round(2)
print(summary_stats)
```

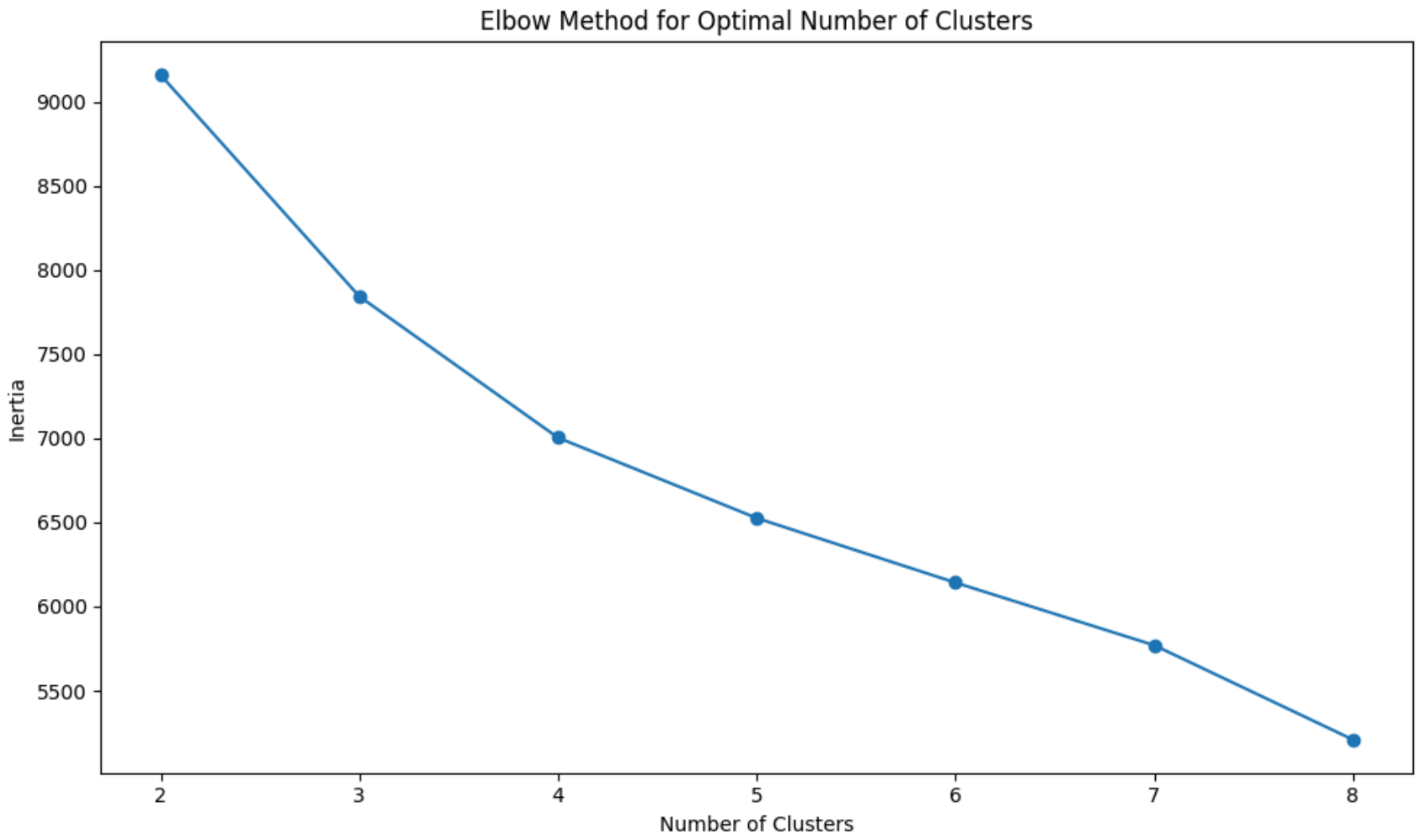
```
          Income      Age  Recency  Total_Spending  Total_Purchases  \
count    2213.00  2213.00  2213.00         2213.00          2213.00
mean    52236.58    57.08    49.01          607.02            12.56
std     25178.60    11.70    28.94          602.49             7.20
min      1730.00    30.00     0.00            5.00             0.00
25%     35246.00    49.00    24.00           69.00             6.00
50%     51373.00    56.00    49.00          397.00            12.00
75%     68487.00    67.00    74.00         1048.00            18.00
max    666666.00    86.00    99.00         2525.00            32.00

       NumWebVisitsMonth  Total_Children  Response
count            2213.00         2213.00   2213.00
mean                5.32            0.95      0.15
std                 2.43            0.75      0.36
min                 0.00            0.00      0.00
25%                 3.00            0.00      0.00
50%                 6.00            1.00      0.00
75%                 7.00            1.00      0.00
max                20.00            3.00      1.00
```

Clustering

In [21]:
```
cluster_features = [
    "Income",
    "Recency",
    "Total_Spending",
    "NumWebPurchases",
    "NumCatalogPurchases",
    "NumStorePurchases",
    "NumWebVisitsMonth"
]
```

```
cluster_df = df[cluster_features].copy()

# Scale data

scaler = StandardScaler()
cluster_scaled = scaler.fit_transform(cluster_df)
```

In [22]:
```
# Use elbow method to compare inertia across cluster counts
inertia_vals = []
cluster_range = range(2, 9)

for k in cluster_range:
    kmeans_model = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans_model.fit(cluster_scaled)
    inertia_vals.append(kmeans_model.inertia_)

plt.figure(figsize=(10, 6))
plt.plot(cluster_range, inertia_vals, marker="o")
plt.title("Elbow Method for Optimal Number of Clusters")
plt.xlabel("Number of Clusters")
plt.ylabel("Inertia")
plt.tight_layout()
plt.show()
```



In [23]:
```
# Compare silhouette scores for possible cluster counts
silhouette_vals = []
cluster_range = range(2, 9)

for k in cluster_range:
    kmeans_model = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans_model.fit_predict(cluster_scaled)
    score = silhouette_score(cluster_scaled, labels)
    silhouette_vals.append(score)

silhouette_results = pd.DataFrame({
```

```
    "Clusters": list(cluster_range),
    "Silhouette_Score": silhouette_vals
})

print(silhouette_results)
```

```
   Clusters  Silhouette_Score
0         2          0.383526
1         3          0.307341
2         4          0.233066
3         5          0.222931
4         6          0.219361
5         7          0.219832
6         8          0.224236
```

In [24]:

```
# Fit final KMeans model
final_k = 4
kmeans_final = KMeans(n_clusters=final_k, random_state=42, n_init=10)
df["Cluster"] = kmeans_final.fit_predict(cluster_scaled)

# Check cluster counts
print(df["Cluster"].value_counts().sort_index())
```

```
Cluster
0    557
1    538
2    586
3    532
Name: count, dtype: int64
```

In [25]:

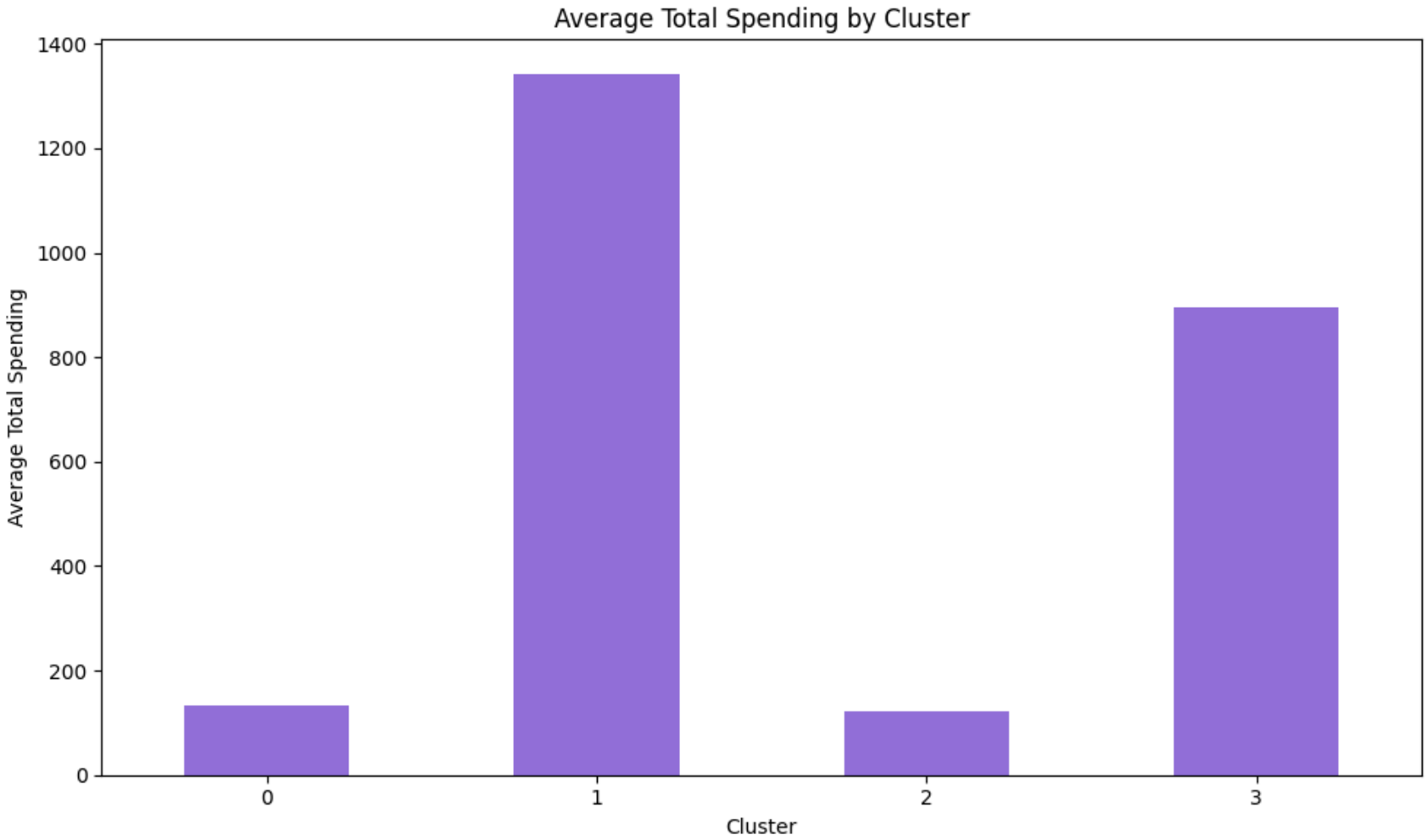
```
# Profile the clusters using average values
cluster_profile = df.groupby("Cluster")[[
    "Income",
    "Recency",
    "Total_Spending",
    "Total_Purchases",
    "NumWebVisitsMonth",
    "Response",
    "Total_Children",
    "Age"
]].mean().round(2)
```

```
print(cluster_profile)
```

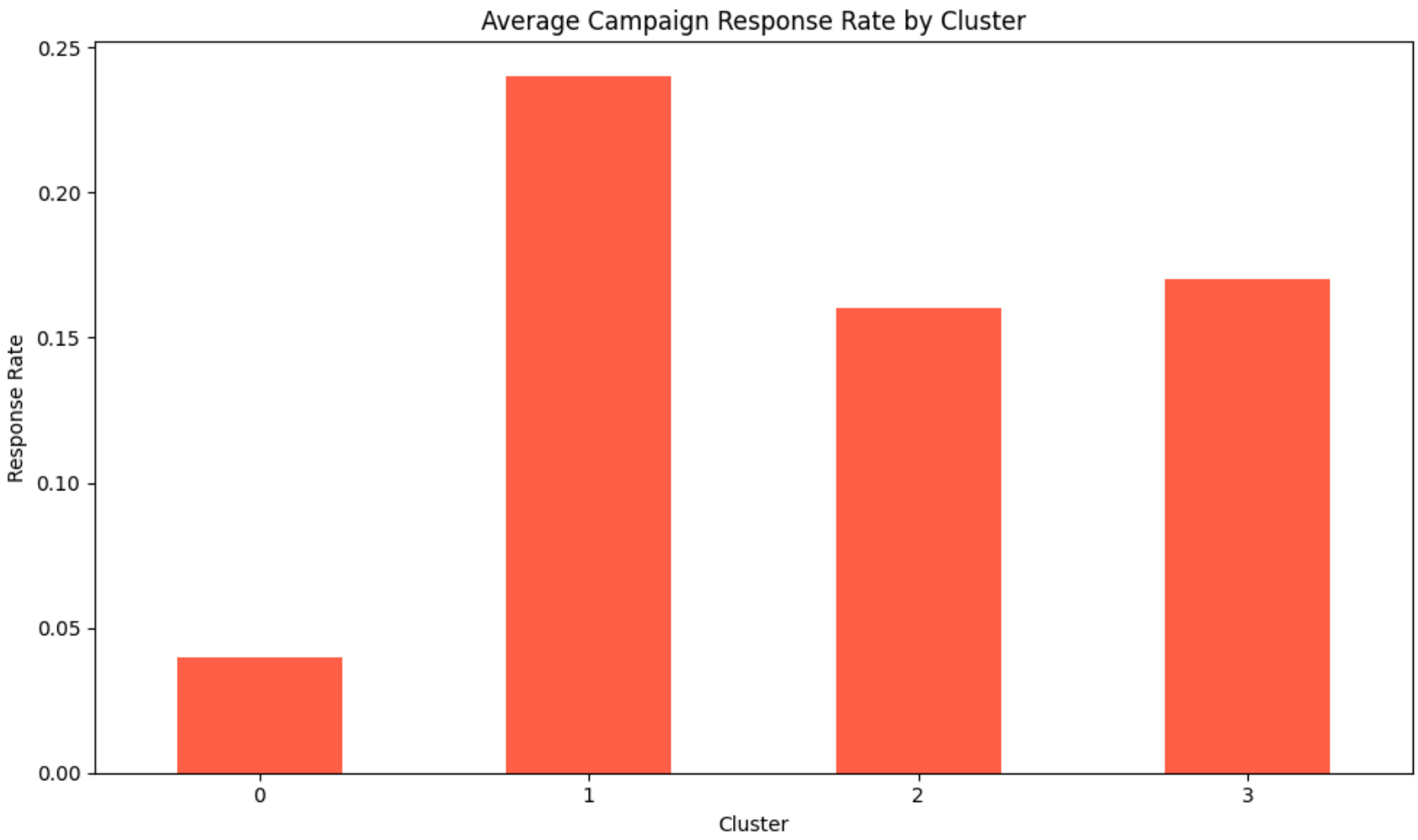
```
           Income  Recency  Total_Spending  Total_Purchases  \
Cluster
0        36938.64    74.98          133.22             6.74
1        78165.75    50.68         1341.26            18.96
2        34818.09    24.17          121.08             6.37
3        61218.34    47.48          895.84            19.02

         NumWebVisitsMonth  Response  Total_Children    Age
Cluster
0                     6.22      0.04            1.26  56.01
1                     2.39      0.24            0.27  58.56
2                     6.64      0.16            1.21  54.61
3                     5.89      0.17            1.01  59.43
```

In [26]:

```
# Plot: Average total spending by cluster
plt.figure(figsize=(10, 6))
cluster_profile["Total_Spending"].plot(kind="bar", color="mediumpurple")
plt.title("Average Total Spending by Cluster")
plt.xlabel("Cluster")
plt.ylabel("Average Total Spending")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

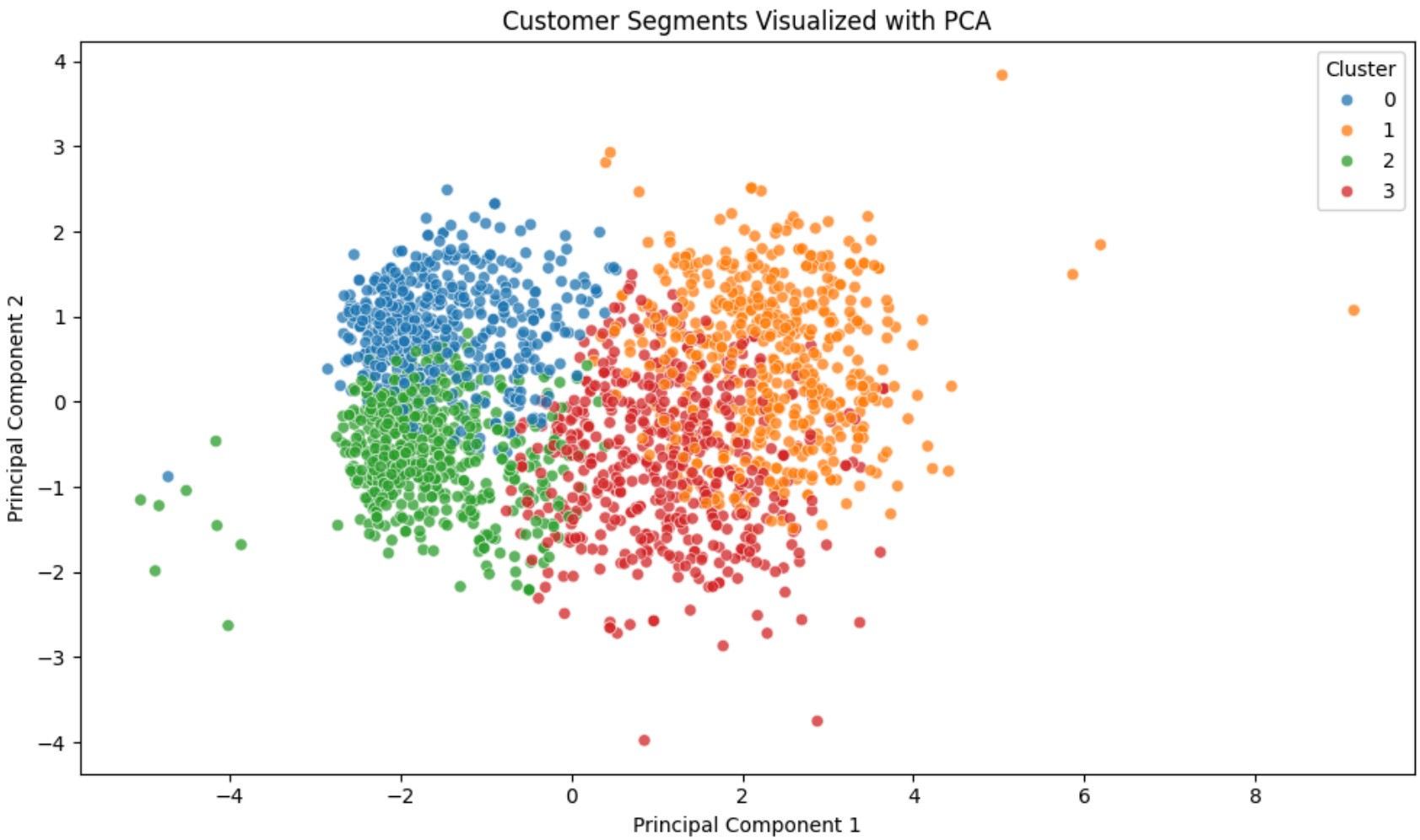


In [27]:
```
# Plot: Campaign response rate by cluster
plt.figure(figsize=(10, 6))
cluster_profile["Response"].plot(kind="bar", color="tomato")
plt.title("Average Campaign Response Rate by Cluster")
plt.xlabel("Cluster")
plt.ylabel("Response Rate")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



In [28]:
```
# Use PCA to reduce the clustering features to 2 dimensions for visualization
pca_model = PCA(n_components=2)
pca_components = pca_model.fit_transform(cluster_scaled)

pca_df = pd.DataFrame(pca_components, columns=["PC1", "PC2"])
pca_df["Cluster"] = df["Cluster"].values

# Plot 5: PCA projection of customer clusters
plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=pca_df,
```

```
    x="PC1",
    y="PC2",
    hue="Cluster",
    palette="tab10",
    alpha=0.75
)
plt.title("Customer Segments Visualized with PCA")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```

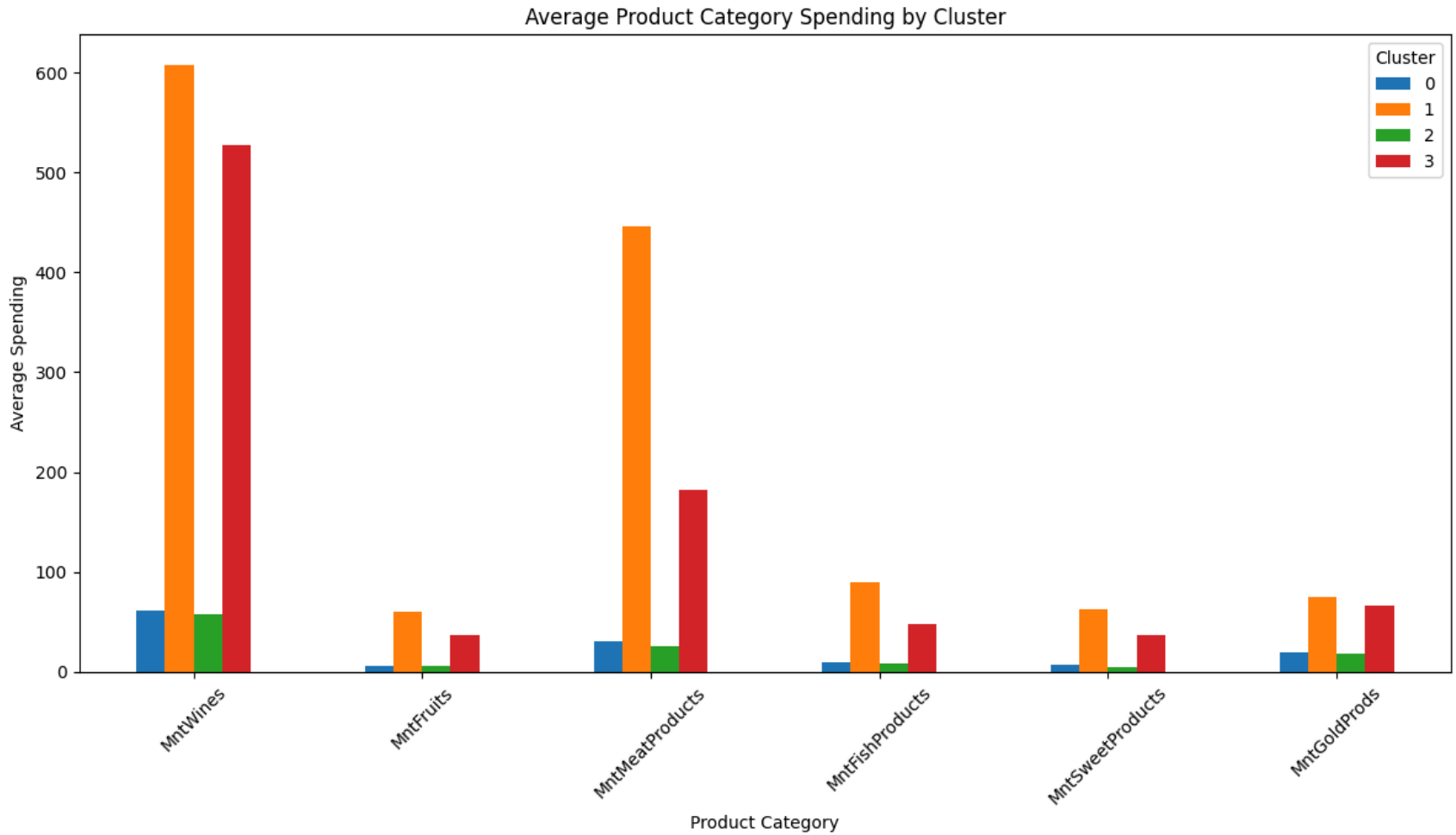


In [29]:
```
# Examine spending by product category across clusters
category_profile = df.groupby("Cluster")[spend_cols].mean().round(2)
print(category_profile)
```

```
         MntWines  MntFruits  MntMeatProducts  MntFishProducts  \
Cluster
0           60.91       6.30            30.44             9.14
1          607.65      60.06           446.35            89.91
2           57.96       5.48            26.10             8.08
3          527.25      36.13           182.52            47.15

         MntSweetProducts  MntGoldProds
Cluster
0                    6.75         19.68
1                   62.50         74.79
2                    5.09         18.36
3                   36.57         66.21
```

In [30]:

```
# Plot: Product spending profile by cluster
category_profile.T.plot(kind="bar", figsize=(12, 7))
plt.title("Average Product Category Spending by Cluster")
plt.xlabel("Product Category")
plt.ylabel("Average Spending")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [31]:
```
# Save cleaned dataset with cluster labels for appendix or export
df.to_csv("customer_segmentation_clustered_output.csv", index=False)

# Save cluster profile table
cluster_profile.to_csv("cluster_profile_summary.csv")

# Save category profile table
category_profile.to_csv("cluster_category_profile.csv")

print("Files saved successfully.")
print("customer_segmentation_clustered_output.csv")
```

```python
print("cluster_profile_summary.csv")
print("cluster_category_profile.csv")
```

```
Files saved successfully.
customer_segmentation_clustered_output.csv
cluster_profile_summary.csv
cluster_category_profile.csv
```